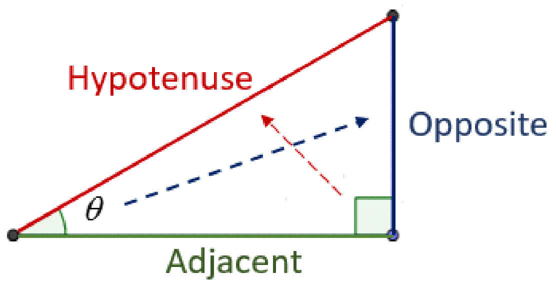
Week 3 - Lesson 3 - Cosine Ratio

Wednesday, 8 April 2020 10:44 AM

Handy saying to help you know which ratio to use: **SOH CAH TOA**

SOHCAHTOA



SOH $\quad \text{Sin}\,\theta = \dfrac{\text{Opposite}}{\text{Hypotenuse}}$

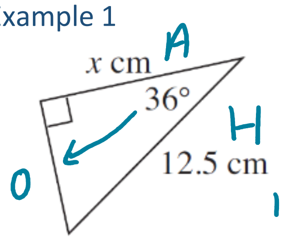
CAH $\quad \text{Cos}\,\theta = \dfrac{\text{Adjacent}}{\text{Hypotenuse}}$

TOA $\quad \text{Tan}\,\theta = \dfrac{\text{Opposite}}{\text{Adjacent}}$

IMPORTANT

Cosine ratio: $\quad \cos(\theta) = \dfrac{\text{adjacent}}{\text{hypotenuse}}$

Example 1



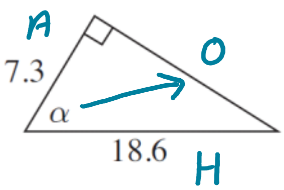
$$\cos\theta = \frac{A}{H}$$

$$12.5 \times \cos 36^\circ = \frac{x}{\cancel{12.5}} \times \cancel{12.5}$$

$$x = 12.5 \times \cos 36^\circ$$

$$x = 10.11\text{cm}$$

Example 2



SOH CAH TOA

$$\cos\theta = \frac{A}{H}$$

$$\cos\theta = \frac{7.3}{18.6} = 0.387$$

$$\cos^{-1}(7.3 \div 18.6) \qquad \cos^{-1}(0.387)$$

$$\theta = 67.2^\circ \qquad \theta = 67.2^\circ$$